

6. Radioactive isotopes decay at different rates.
The decay rate depends on the half-life of a particular radioisotope and the number of radioactive atoms remaining.

(a) State what is meant by the term half-life.

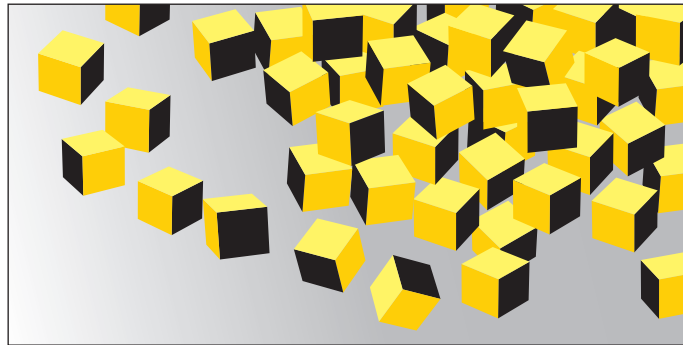
[2]

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- (b) The decay of a radioactive isotope can be modelled using 60 cubes.
Each cube has **one face** shaded black.
These cubes are shown in the picture below and are called **1F** cubes.



Describe the method used to collect results to find the half-life.

[3]

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- (c) In a class experiment, 12 groups of students are each given 60 of these **1F** cubes. The table below shows the results from one group and for the entire class.

Throw	Number of 1F cubes remaining	
	For the group	For the class
0	60	720
1	50	600
2	42	500
3	36	420
4	29	350
5	24	290

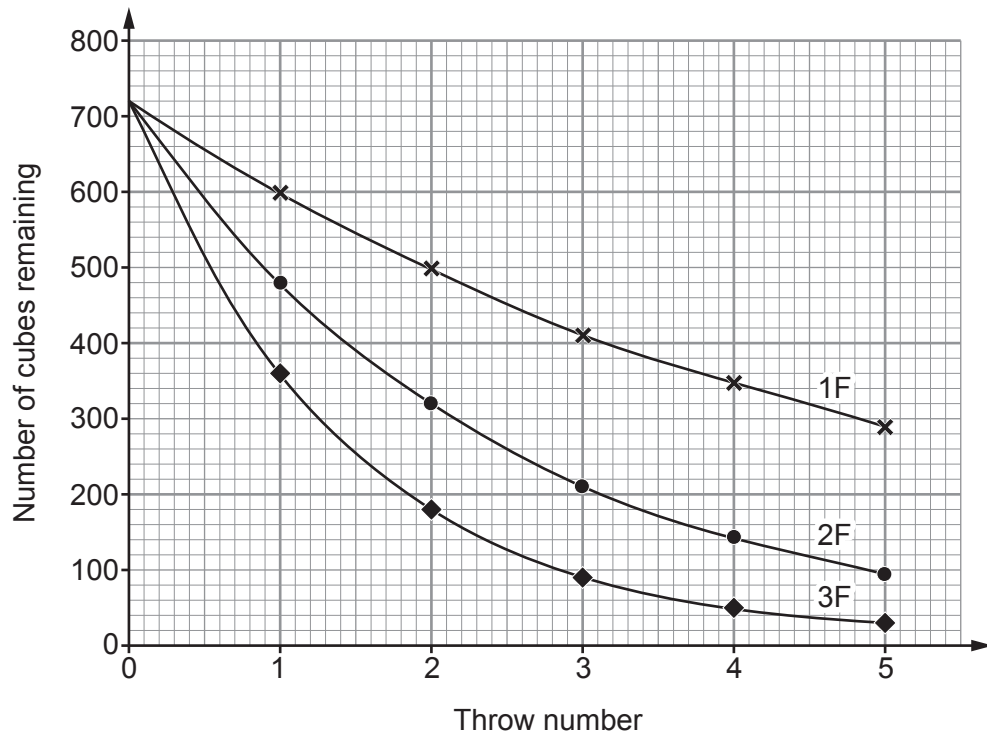
- (i) The results of all the groups were added to give a class result. State why the class result is used to plot a graph and not just the result for one group. [1]

- (ii) The class repeats the experiment with different cubes. The **2F** cubes have **two faces** shaded black and the **3F** cubes have **three faces** shaded black. The class results are shown below.

Throw	Number of cubes remaining in the class	
	For 2F cubes	For 3F cubes
0	720	720
1	480	360
2	320	180
3	210	90
4	144	45
5	95	23



The class results for **1F**, **2F** and **3F** cubes are plotted on the grid below.



- I. **Add lines** to the graph to determine the half-life of the **2F** cubes. [2]

half-life = throws

- II. Use the information in the graphs or tables to tick (✓) the boxes next to the **three** correct statements. [3]

1F cubes show decay at the slowest rate

☐

2F cubes show decay of the quickest rate

☐

3F cubes show the equivalent of two half-lives after 2 throws

☐

3F cubes have the longest half-life

☐

1F cubes could also have been replaced with 720 coins and the number of 'heads' counted after each throw

☐

The number of **2F** cubes remaining after 4 throws is about $\frac{1}{5}$ of the original

☐

TURN OVER FOR THE LAST PART OF THE QUESTION



- (d) Sodium-24 is a radioactive isotope with a half-life of 15 hours.
A solution containing 48.0 mg of sodium-24 was injected into a patient for a medical investigation.

Calculate the mass of sodium-24 that will be left after 75 hours.

[3]

mass = mg

END OF PAPER

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